

INTRODUCTION

Several interdependent decisions are involved in the formulation of a marketing strategy for a brand (of a product or service). These include decisions on not only the product's characteristics but also its positioning, communication, distribution, and pricing to chosen sets of targeted customers. The decisions will need to be made in the wake of uncertain competitive reactions and a changing environment. For a business to be successful, the decision process must include a clear understanding of how customers will choose among (and react to) various competing alternatives. It is well accepted in marketing that choice alternatives can be described as profiles on multiple attributes and that individuals consider various attributes while making a choice. While choosing, consumers typically make trade-offs among the attributes of a product or service. Conjoint analysis (CA) is a set of techniques ideally suited to studying customers' choice processes and determining trade-offs.

CA is probably the most significant development in marketing research methodology over the last 40 years or so. The method has been applied in several thousand applied marketing research projects since its introduction to the marketing researchers in 1971 (Green and Rao, 1971). The method has been applied successfully for tackling several marketing decisions such as optimal design of new products, target market selection, pricing a new product, and competitive reactions. A significant advantage of the method has been the ability to answer various "what if" questions using market simulators; these simulators are based on the results from a conjoint study for hypothetical and real choice alternatives.¹

Five different features of CA have contributed to its versatility for tackling marketing managerial problems: (i) it is a measurement technique for quantifying buyer trade-offs and values; (ii) it is an analytical technique for predicting buyers' likely reactions to new products/services; (iii) it is a segmentation

technique for identifying groups of buyers who share similar trade-offs/values; (iv) it is a simulation technique for assessing new product service ideas in a competitive environment; and (v) it is an optimization technique for seeking product/service profiles that maximize share/return (Green, Krieger, and Wind, 2004).

Against this brief background, this article will be organized as follows. In the next (second) section, principal types of CAs that are in vogue in marketing research are described. In the third section, the process for conducting a conjoint study is briefly described; this section also includes a discussion of various data collection formats and designs for developing stimuli (or profiles) for a conjoint research problem. The basics of conjoint models and estimation are described in the fourth section and a simplified illustration of one approach is provided. In the fifth section, an overview of the variety of applications of this method is presented. A series of recent developments and future directions are enumerated in the sixth section with limited elaboration.

PRINCIPAL TYPES OF CONJOINT ANALYSIS

Over the past several years, various researchers have contributed to the general methodology of CA. The reader is referred to Green and Srinivasan (1978, 1990) for excellent reviews of the field of CA. Essentially, there are four types of conjoint methods; the traditional method (CA) that uses stated preference ratings; the choice-based conjoint analysis (CBCA) that uses stated choices; the adaptive conjoint analysis (ACA) developed in part to handle the issue of a large number of attributes; and the self-explicated CA, which is a bottom-up method. The first three of these can be called *decompositional methods* because the stated preference or stated choice data are decomposed to obtain part-worth functions. The fourth one is called the *compositional method* because it composes a preference score from ratings of scores on attribute levels and relative importances of attributes. We briefly describe each of these.

The traditional CA collects preferences (judgments) for profiles of hypothetical products each described on the entire set of attributes

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selected for the conjoint study. These profiles are called *full profiles*. However, when one concatenates levels of all attributes, the complete set of full profiles (or full factorial design), will in general, be very large. A respondent will be unduly burdened when asked to provide preference judgments on all profiles. Typically, a smaller set of full profiles (selected according to an experimental design) is used in a conjoint study. An individual's overall stated preferences are decomposed into separate and compatible utility values corresponding to each attribute typically using regression-based methods. These separate functions are called *attribute-specific part-worth functions*. In most cases, the preference functions can be estimated at the individual level. This estimated preference function can be deemed as an indirect utility function.

While the traditional decompositional conjoint approach involved full profiles of product concepts described on multiple attributes, several new data collection formats have emerged over the years. A significant development is the use of data on stated choices elicited under hypothetical scenarios that mimic the marketplace and estimating part-worth functions from such data using primarily multinomial logit methods (see LOGIT MODEL); these methods are labeled choice conjoint methods (CBCA or CBC) and have become popular in the early 1990s and are probably the most widely used methods currently. They are based on the behavioral theory of random utility maximization (McFadden, 1974; the origin of this approach is the law of comparative judgment developed by Thurstone (1927)). This approach decomposes an individual's random utility for an object into two parts: deterministic utility and random part. Depending on the distributional assumptions for the error part, a number of alternative models are developed to describe the probability of choice of an object. The most popular one is the multinomial logit model (MNL) (see LOGIT MODEL) that uses the extreme value distribution for the error term. These methods belong to the family of discrete choice analysis methods. An excellent volume that elaborates on these stated choice methods is by Louviere, Hensher, and Swait (2000); see also Ben-Akiva and Lerman (1991).

Researchers have also developed an adaptive conjoint method which is called *ACA* (Johnson, 1987). The method involves first a self-explicated task (i.e., eliciting data on attribute importances and attribute level desirabilities using ranking and subsequent rating) followed by preference ratings for a set of partial profiles descriptions, two at a time using a graded, paired comparison scale. The partial profile descriptions are tailored to each respondent based on the data collected in the self-explicated task. Both the tasks are administered by computer. This method is a type of hybrid² model approach.

In contrast, the compositional approach based on the multiattribute attitude models (Wilkie and Pessemier, 1973) estimates preferences from judged values of the components (importances and desirabilities) that contribute to preference. In the compositional approach, individuals are asked to evaluate the desirability of each level of all the attributes as well as the relative importances assigned to the attributes. Then, the preference for any product concept is estimated as a weighted sum of the desirabilities for the specific levels of attributes describing that concept; the weights are the relative importances. This approach is called the "*self-explicated*" method (see Green and Srinivasan, 1978 for more details). Studies have shown that the self-explicated method is surprisingly quite robust (Srinivasan and Park, 1997).

PROCESS OF CONDUCTING A CONJOINT STUDY

In order to provide a context for designing a conjoint study, consider the problem of determining the steady-state demand for a new product. Assume that a sample of n consumers drawn randomly from a total of N customers in the target market for this product was interviewed in a conjoint study. Let q_i denote the quantity of product bought by i th customer in the sample; $i = 1, 2, \dots, n$ (generally measured in the survey) and let p_i denote the probability that the i th customer will purchase the new product in a steady state (conditional on his/her consideration set of alternative items including the new product). Then, the demand forecast

for the new product is given by the model:

$$D = \left(\frac{N}{n}\right) \sum_{i=1}^n q_i p_i$$

The problem then is to estimate the probability of purchase p_i for the new product for the members of the sample. There are at least two solutions for this problem.

One solution is to employ the traditional stated preference-based CA. This method involves obtaining an estimate of the utility a customer derives for a new product described in terms of its attributes relative to other items considered and then translating the utility into probabilities of purchase. Several methods exist for this transformation; see Green and Krieger (1988). The second solution is to utilize the CBCA method. In this approach, the probabilities can be computed directly from the MNL model. Against this backdrop, the process of designing conjoint studies is reviewed.

A typical CA project for collecting and analyzing stated preference or stated choice data³ as such consists of four main steps: (i) development of stimuli based on a number of salient attributes (hypothetical profiles or choice sets); (ii) presentation of stimuli to an appropriate sample of respondents; (iii) estimation of part-worth functions for the attributes as well as any heterogeneity among the respondents; and the use of the estimates in tackling any managerial problems (e.g., forecasting, pricing, or product design). The steps are schematically shown in Figure 1.

One major step is the design of stimuli (either profiles or choice sets). To illustrate profiles and choice sets, consider a simple conjoint problem with three attributes, A, B, and C, each described at four levels, as a1, a2, a3, and a4. An example profile is (a2, b3, c1) and an example choice set is {(a1, b2, c3); (a2, b1, c4); (a4, b3, c2); (No choice)} with some times “no choice” not included. Stated preference for a profile is measured as a rating or a rank relative to other profiles, while stated choice is the choice made by the respondent among the items in a choice set.

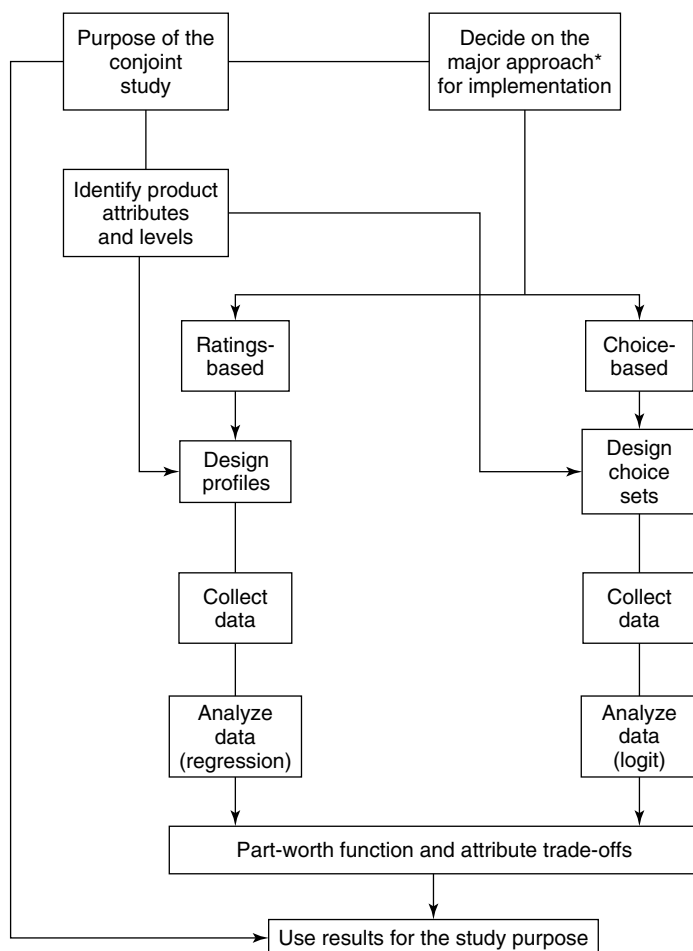
The aspect of designing stimuli (profiles or choice sets) has received much focus since the beginning of CA; it draws much from the theory

of experimental design, where procedures for constructing subsets of combinations of all attribute levels are developed. CA for ratings-based studies makes extensive use of orthogonal arrays (Addelman, 1962; Green, 1974). The process for designing choice sets for collecting stated choice data is a lot more complicated; after developing a number of profiles (usually a subset of all possible profiles), subsets of profiles (4 or 5) are used as choice sets. The “no choice” option is generally included in each choice set. Street and Burgess (2007) present the theory and methods for the construction of optimal stated choice experiments; see also Street, Burgess, and Louviere (2005) and Street and Burgess (2004). Researchers can use the OPTEX procedures in the SAS system (2002–2003) for designing profiles or choice sets; see also Kuhfeld (2005).

In the ratings-based conjoint approach, the respondent is given a number of profiles of product concepts, each described on the attributes under study, and is asked to state his/her preference for each profile on a rating scale (e.g., 10 points or 100 points). These preference data are analyzed using multiple regression methods (typically, a dummy variable ordinary least squares (OLS) regression) to estimate a utility function for each respondent (or for a subgroup of respondents). We illustrate this approach in the next section. Typically, additive utility functions are used although utility functions with interaction terms are possible depending on the experimental designs used for constructing profiles.

The attributes in a conjoint study are of two types: categorical or continuous (or interval-scaled) with only a few selected values (see MODELS FOR CATEGORICAL DEPENDENT VARIABLES). A categorical attribute (such as low, medium, or high) can be converted into a number of dummy variables (one less than the number of levels). A continuous attribute (such as price of a product) can be used directly or can also be converted into dummy variables; if used directly, only a linear term or with both linear and quadratic terms to account for any nonlinear effects can be included in the utility function. With suitable redefinitions of variables, the utility function for the ratings methods can be written as $y = X\beta + \varepsilon$, where

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*Several alternatives exist here; two are highlighted.

Source: Rao, V. R. "Developments in conjoint analysis" in B. Wierenga (ed.) *Handbook of marketing decision models*, New York: Springer science + business media, LLC, 2008.

Figure 1 Major steps in a conjoint study (reproduced from B. Wierenga (Ed) *Handbook of Marketing Decision Models*, © Springer Science + Business Media, LLC, 2008).

ε is the random error of the model assumed to be normally distributed with zero mean and variance of σ^2 , y is the rating on given profile, and X is the corresponding set of p dummy (or other) variables. The model is estimated using regression methods (usually ordinary least squares method; see MULTIPLE REGRESSION). The β is a $p \times 1$ vector of regression coefficients associated with the dummy variables or continuous variables included in the model.

The part-worth values for each attribute can be derived from these regression coefficients.

In the choice conjoint methods, the respondent is given a number of choice sets, each choice set consisting of a small number (typically 4 or 5) of profiles, and is asked to indicate which profile will be chosen. An MNL is used for estimating the deterministic component of the random utility using maximum likelihood methods. A variety of extensions and alternatives exist for

analyzing stated choice data. The MNL model (see LOGIT MODEL) for the choice conjoint data is the probability of choosing profile j in choice set $C = \exp(-v_j) / \sum \exp(-v_k)$, where the summation is taken over all the profiles in the choice set C and v_j is the deterministic component of the utility for the profile j . The deterministic utility function v is specified analogous to the linear combination to the function for y in the ratings methods. The estimated coefficients will be used in computing the part-worth values for the attributes in the study.

Current approaches for implementing a CA project differ in terms of several features; some main features are stimulus representation, formats of data collection, nature of data collection, and estimation methods. Table 1 lays out some alternatives for these features. There is no clear agreement as to which data collection format is the best and all the formats shown in Table 1 are in vogue.

One notable development is the use of hierarchical Bayesian estimation methods (see UNOBSERVED HETEROGENEITY) which enable an

analyst to incorporate prior knowledge in the part-worth values as monotonic or other types of order constraints in the estimation process (Allenby, Arora, and Ginter, 1995); see also Lenk *et al.* (1996). Further, part-worth functions are estimated at the aggregate (or subgroup) level or at an individual level. Researchers have also used finite mixture methods (DeSarbo *et al.*, 1992) to “uncover” segments of respondents based on the preference or choice data collected in conjoint studies; see also Andrews, Ansari, and Currim (2002). The variety of recently developed techniques for estimation of part-worth functions is very impressive and a discussion of these is beyond the scope of this article. For a recent discussion of conjoint methods, see Hauser and Rao (2004) and Rao (2008).

BASICS OF CONJOINT MODELS

Conjoint methods are intended to “uncover” the underlying preference function of a product in terms of its attributes⁴. A general product profile defined on r attributes can be written as

Table 1 Alternatives for selected features of conjoint analysis.

<i>Representation of Stimuli</i>	<i>Formats of Data Collection</i>	<i>Nature of Data Collection</i>	<i>Estimation Methods</i>
Verbal descriptions	Full profile evaluations	One-shot Adaptive	Regression-based methods
Pictorial descriptions	Partial profile evaluations	Multiple times ^a	Random utility models
Videotapes and supporting materials	Stated preferences		Direct computation based on self-explicated importances
	Stated choices		Hierarchical Bayes estimation ^a
Virtual prototypes	Self-explicated methods		
Combinations of physical models, photographs and verbal descriptions	Configurators ^a		Methods based on new optimization methods ^a Analytic center estimation, support-vector machines, genetic algorithms

^aThese are newer methods.
Source: Adapted from Hauser and Rao (2004) and Rao (2008).

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$(x_{j1}, x_{j2}, \dots, x_{jr})$, where x_{jt} is the level for the j th profile on the t th attribute in a product profile. While there exist several ways for specifying the preference functions in CA, researchers usually start with an additive conjoint model; but, the theory extends to models with interactions as well. The preference score⁵ for the j th product profile, y_j for one respondent additive conjoint model is

$$y_j = U_1(x_{j1}) + U_2(x_{j2}) + \dots + U_r(x_{jr})$$

where $U_t(\bullet)$ is the component utility function specific to the t th attribute (also called *part-utility function* or *part-worth function*). No constant term is specified, but it could be included in any one of the U -functions or assumed to be zero (without any loss of generality). The specification of the U -function for any attribute will depend upon its type (categorical and quantitative). In practice, a conjoint study may contain both types of attributes.

Brand names or verbal descriptions such as high, medium, or low are examples of a categorical attribute; here the levels of the attribute are described by words. A quantitative attribute is one measured by either an interval scale or a ratio scale; numbers describe the “levels” of such an attribute; examples are the weight of a laptop and speed of the processor.

The levels of a categorical attribute can be recoded into a set of dummy variables and a part-worth function is specified as a piecewise linear function in the dummy variables. In this case, the component utility function for a categorical attribute (t th for example) will be

$$U_t(x_{jt}) = D_{t1}U_{t1} + D_{t2}U_{t2} + \dots + D_{tr_t}U_{tr_t}$$

where r_t is the number of discrete levels for the t th attribute (resulting from the construction of the profiles or created ex post); D_{tk} is a dummy variable taking the value 1 if the value x_{it} is equivalent to the k th discrete level of x_t and 0 otherwise; and U_{tk} is the component of the part-worth function for the k th discrete level of x_t . In practice, only $(r_t - 1)$ – one less the number of discrete levels of the attribute – dummy variables are necessary for estimation.

A quantitative attribute can be used in a manner similar to a categorical attribute by

coding its values into categories or used directly in the specification of the part-worth function for the attribute. In the latter case, the function can be specified as linear (vector model) or nonlinear; one example of a nonlinear function is the ideal point model. Mathematically, the component utility function can be specified as

$$U_t(x_{jt}) = \begin{cases} w_t x_{jt} & \text{for the vector model; and} \\ w_t (x_{jt} - x_{0t})^2 & \text{for the ideal point model;} \end{cases}$$

where w_t is a weight (positive or negative) and and x_{0t} is the ideal point on the t th attribute.

A linear function is appropriate for an attribute deemed to be desirable (e.g., speed of a laptop computer) or undesirable (e.g., weight of a laptop computer); such a function is called a vector model for which the utility increases (or decreases) linearly with the numerical value of the attribute.

As mentioned above, with suitable redefinitions of variables, the preference function can be written as $y = X\beta + \varepsilon$, where ε is the random error of the model assumed to be normally distributed with zero mean and variance of σ^2 ; y is the rating on a given profile; and X is the corresponding set of p dummy (or other) variables. The β is a $p \times 1$ vector of part-worths among the levels of attributes.

At this point, it will be useful to indicate the software available for designing and implementing conjoint studies. These are

- Sawtooth Software (ACA, CBC, etc.; probably the most complete solution)
- SPSS (useful for preference-based approach)
- SAS (OPTEX for design and several other programs for analysis)
- LIMDEP (useful for analyzing data of various types; Greene, 2003)
- Bayesm package in R (developed by Rossi, Allenby, and McCulloch, 2005)
- MATLAB (one needs to develop a specific program code).

Table 2 Variable smart phone attributes.

Attribute	Levels			
	1	2	3	4
Style	Candy bar 	Slide phone 	Flip phone 	Touch screen 
Brand	Blackberry	Nokia	LG	Samsung
   				
Talk time	3 h	5 h	7 h	9 h
Weight	100 g	115 g	130 g	145 g
Camera quality	2 MP	3 MP	6 MP	8 MP

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AN ILLUSTRATION OF RATINGS-BASED
CONJOINT ANALYSIS

As an illustration of the ratings-based conjoint method, assume that a wireless provider firm is interested in determining trade-offs among various features of a smart phone (a technologically advanced product with a number of features). In order to simplify the data collection, assume that the firm is interested in the trade-offs among five attributes, namely, style of the phone, brand name, talk time, weight, and camera quality (having predetermined a number of standard features). Price attribute was not included because it was part of a contract with the wireless provider and was about the same for all brands. Each of these five attributes is varied at four levels. Table 2 shows the features that are predecided and the levels of the five features varied in the study.

The total number of possible hypothetical profiles are 1028 ($= 4 \times 4 \times 4 \times 4 \times 4$), which are combinations of the levels of the five attributes. Given that it is almost impossible to have a respondent judge all these profiles, the study designer has selected 32 of these profiles using a fractional factorial design. In the study, respondents were shown the complete list of standard features and were asked to provide preferences on a 0–100 point scale for the 32 profiles. These profile descriptions were provided using a computerized questionnaire. The results from an analysis of one respondent's evaluations are shown. These data are analyzed using dummy variable regression after converting each attribute into three dummy variables as shown in Table 3 to obtain part-worth functions for the five attributes. The resulting regression and part-worth values are also shown in Table 3. The range of each part-worth function is a simple measure⁶ of the importance of that attribute. Figure 2 shows the plots of part-worth functions for the five attributes.

Although not shown, the fit of the part-worth model to the individual's preference ratings is quite good with an adjusted R-square of 0.88. On the basis of this analysis, one can conclude that this respondent has a strong preference for a flip style Blackberry smart phone that is lightest in weight with a talk time of 9 hours and a camera quality of 6 MP. From the graphs

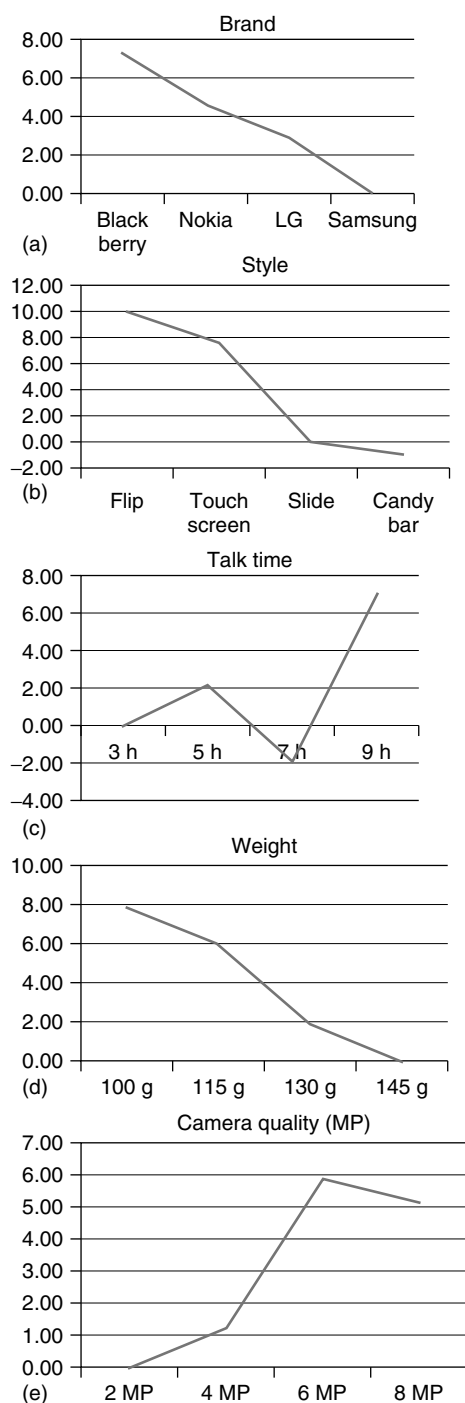


Figure 2 Part-worth functions for the smart phones illustration weight.

Table 3 Estimated part-worth values for the illustrative data for smart phone study.

Attribute	Level	Recoded Dummy Variables			Part-worth Value	Relative Importance(%)
		D1	D2	D3		
Brand	Blackberry	1	0	0	7.25	17.86
	Nokia	0	1	0	4.63	
	LG	0	0	1	2.87	
	Samsung	0	0	0	0.00	
Style	Candy bar	1	0	0	-0.97	26.42
	Flip	0	1	0	9.95	
	Touch screen	0	0	1	7.58	
	Slide	0	0	0	0.00	
Talk time	3 h	0	0	0	6.99	21.84
	5 h	0	0	1	-1.88	
	7 h	0	1	0	2.12	
	9 h	1	0	0	0.00	
Weight	100 g	1	0	0	7.88	19.4
	115 g	0	1	0	6.01	
	130 g	0	0	1	1.89	
	145 g	0	0	0	0.00	
Camera quality	2 MP	0	0	0	0.00	14.48
	4 MP	1	0	0	1.22	
	6 MP	0	1	0	5.88	
	8 MP	0	0	1	5.15	

of the part-worths, one can see that the decline in utility from these levels of the attributes to other levels is not uniform. Further, there is nonlinearity in the part-worth function for the attribute of the camera quality. Looking at the relative importances, this individual places most importance for the style attribute followed by talk time, weight, brand, and camera quality in that order. The part-worth functions can be used to predict the individual's preference rating for a profile not covered in the 32 profiles. Further, one can estimate preferences for items in any choice set; these estimates can be used to predict the individual's first choice and other choices. Also, if the study is conducted for a sample of respondents, the vector of estimated relative importances can be used to form clusters of individuals whose importances are quite similar; these clusters are akin to market segments. Focusing on one brand (e.g., LG), one can make predictions of first and other choices for the sample for various scenarios (e.g., anticipated changes in the product designs of competing

brands); such a process is the simulation aspect of CA, which is highly useful for managers.

APPLICATIONS

Since its introduction, conjoint methods have been applied in a large number of applied marketing research projects. There is no recent estimate of the number of applied studies but its use is increasing tremendously. The conjoint methodology has been applied in several areas; these include consumer nondurable products (bar soaps, carpet cleaners, lawn chemicals, etc.), industrial goods (copying machines, portable computer terminals, personal computer design, etc.), other products (car batteries, ethical drugs, pesticides, etc.), financial services (branch bank services, auto insurance policies, credit card features, etc.), transportation (domestic airlines, electric car design, etc.), and other services (hotel design, car rental agencies, telephone pricing, etc.). The method has been applied

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successfully for tackling several marketing decisions such as the optimal design of new products, target market selection, pricing a new product, and studying competitive reactions. Some high profile applications of these techniques include the development of Courtyard Hotels by Marriott (Wind *et al.*, 1989) and the design of the E-Z Pass Electronic Toll Collection System in New Jersey and neighboring States in the United States (Green, Krieger, and Vavra, 1997). A significant advantage of the conjoint method has been the ability to answer various “what if” questions using market simulators; these simulators are based on the results of an analysis of conjoint data collected on hypothetical and real choice alternatives.

SOME RECENT DEVELOPMENTS

We have mentioned the development of hierarchical Bayesian methods and experimental design described earlier in the article. In addition, there have been developments on dealing with a positive part-function for price (Rao and Sattler, 2003), use of incentive-aligned methods for data collection (Ding, Grewal, and Liechty, 2005; Ding, 2007), a range of methods for handling large number of attributes (reviewed in Rao, Kartono, and Su, 2008), polyhedral methods aimed at reducing respondent burden (Toubia *et al.*, 2003; Toubia, Hauser, and Simester, 2004), and modeling choices for bundles (Bradlow and Rao, 2000; Chung and Rao, 2003) and upgrading methods (Park, Ding, and Rao, 2008) based on the BDM method (Becker, DeGroot, and Marschak, 1964), experimental designs based on new criteria such as utility balance (Huber and Zwerina, 1996; Hauser and Toubia, 2005), continuous CA (Wittink and Keil, 2003; Su and Rao, 2006), adaptive self-explicated analysis (Netzer and Srinivasan, 2007), and measuring reservation prices for single products and bundles (Jedidi and Zhang, 2002; Jedidi, Jagpal, and Manchanda, 2003). These are but only a few examples of continuous developments in CA research. The article written from the 2007 Choice Symposium, Netzer *et al.* (2008) identifies several new directions in this methodology; see also Hauser and Rao (2004), Bradlow (2005), and Rao (2008) for ideas for

future research in this area. In conclusion, one might say that CA is alive and well!

ENDNOTES

¹ It will be useful to review some terms used in conjoint analysis. Attributes are (mainly) physical characteristics that describe a product; levels are the number of different values an attribute takes; profile is a combination of attributes, each attribute at a particular level, presented to a respondent for an evaluation (or stated preference); choice set is a pre-specified number of profiles presented to a respondent to make a pseudo-choice (stated choice).

² Hybrid models involve a combination of several tasks aimed to increase the “efficiency” of data collection in conjoint studies usually for large number of attributes. See Green (1984) for a review of these methods; see also Green and Krieger (1996). We will not delve much into these methods due to space limitations.

³ For the sake of ease in exposition, we will restrict to these two types of data and will not delve into methods that involve variations such as the hybrid methods.

⁴ For an introduction to conjoint analysis, see Orme (2006).

⁵ For exposition purposes, I am considering a ratings-based conjoint analysis where respondents provide preference ratings for a number of product profiles. The same can apply to the v-function in the choice-based conjoint analysis.

⁶ There are several other measures such as partial R-squared but are beyond the scope of this chapter.

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